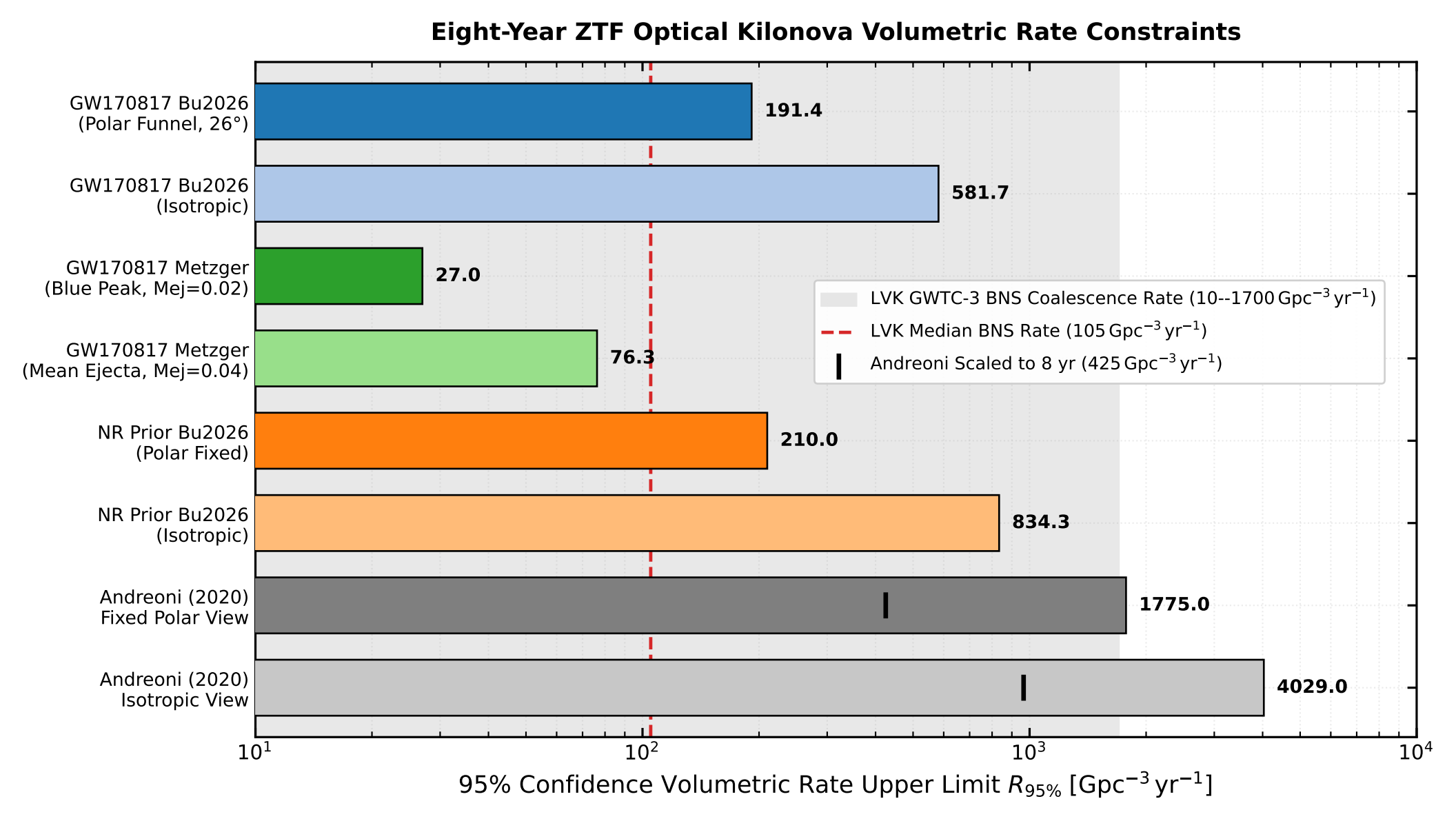
Eight-Year ZTF Optical Kilonova Volumetric Rate Constraints
GW170817 Bu2026
(Polar Funnel, 26°)
191.4
GW170817 Bu2026
(Isotropic)
581.7
GW170817 Metzger
(Blue Peak, Mej=0.02)
27.0
GW170817 Metzger
(Mean Ejecta, Mej=0.04)
76.3
NR Prior Bu2026
(Polar Fixed)
210.0
NR Prior Bu2026
(Isotropic)
834.3
Andreoni (2020)
Fixed Polar View
1775.0
Andreoni (2020)
Isotropic View
4029.0
LVK GWTC-3 BNS Coalescence Rate (10--1700 Gpc^{-3} yr^{-1})
LVK Median BNS Rate (105 Gpc^{-3} yr^{-1})
Andreoni Scaled to 8 yr (425 Gpc^{-3} yr^{-1})
10^1
10^2
10^3
10^4
95% Confidence Volumetric Rate Upper Limit $R_{95\%}$ [Gpc^{-3} yr^{-1}]